

Linear Programming of Graph Squares

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Chapter 1

Introduction

1.1 Historical Background

In 1949, Leon Festinger proposed a novel method in the field of social psychology for analyzing relationships among a number of people present [FBB23]. Classically this was represented using a sociogram, a visual tool that maps out the relationships between people, however Forsyth and Katz introduced the use of a matrix to represent the sociometric pattern, with rows and columns being individuals and containing 1s for choices and blank diagonals. Festinger noticed that the action of squaring a matrix

$$A_{rc}^2 = A_{1c}A_{r1} + \cdots + A_{nc}A_{rn}$$

represented the number of 'two-step' connections, which can then be used to denote 'three-step' connections between any two people

$$A_{rc}^3 = A_{1c}A_{r1}^2 + \cdots + A_{nc}A_{rn}^2,$$

such that higher steps can be easily generated. Evidently, raising matrices to the power generates a group of people where every member is connected to each other - a clique, and identifying such social cliques in a cubed matrix can be done by looking at the diagonals.

This matrix A can be considered as a form of adjacency matrix, and in the process of improving clique detection methods, Ross and Harary showed that complete graphs may be realized as squares of trees by considering powers of such matrices [RH60]. They introduced a structural characterization of the square of a graph in terms of clique intersections, multicliqual vertices, and so called "neighborly" properties, proving that a graph is the square of a tree if and only if

- i) every vertex is "neighborly" - is not isolated,
- ii) no two cliques intersect in more than two vertices, and
- iii) the number of clique pairs intersecting in two vertices is exactly one less than the total number of cliques.

In addition, there must exist a one-to-one correspondence between maximal cliques and multicliqual vertices where a vertex is called multicliqual if it belongs to at least two maximal cliques. If a multicliqual vertex b belongs to exactly k maximal cliques, then the clique corresponding to b contains exactly k multicliqual vertices. Furthermore, if two maximal cliques intersect in exactly one vertex b , then there exists a third maximal clique which contains b and shares exactly one additional vertex with each of the original two cliques. Using these properties, they constructed

an explicit algorithm that recovers the underlying tree by associating multicliqual vertices with maximal cliques and forming a tree from pairwise clique intersections.

Since then, the problem of finding the square root of a graph has become a well-studied field in both mathematics and computer-science, with applications in distributed computing (e.g., see [PA23]), network design and computational biology. Research in this area has primarily focused around an undirected graph H and its square $G = H^2$ which is obtained by adding an edge between every pair of vertices whose distance in H is at most two.

Being the simplest and most straight forward, the two-step undirected graph square has been studied extensively with much of the focus being on the complexity of the decision problem - given a specific square graph G does there exist a root graph H such that $H^2 = G$. Motwani and Sudan showed that graph squares are NP-complete in general and hence subsequent studies in the area have restricted the decision problem in two ways, either by considering various properties on the root graph or by considering properties on the square graph.

1.2 Literature

Focusing first on special undirected graph classes, the aforementioned work by Ross and Harary in 1960 gave an algorithm for constructing a tree root by using characterizations of the square graph. As part of this work they proved tree square roots are unique up to isomorphism. This was the first major contribution to the study of graph roots and provided fundamental structural characteristics of square graphs, laying the groundwork for later research on graph root problems and their applications [RH60].

Fleischer in 1974 showed that as long as a graph is 2-connected its square graph will have a Hamiltonian cycle. This established a broad condition guaranteeing Hamiltonicity in graph squares which is used in many subsequent works [Fle74].

Lau and Corneil in 2004 provided an algorithm for constructing square roots from an interval square graph in polynomial time, and showed that the decision problem for finding roots of chordal graphs, roots of split graphs, and chordal roots of graphs are NP-complete [LC04]. Furthermore, Lau considered the construction of square roots from a bipartite graph in linear time, and showed that the existence of cube roots in bipartite square is NP-complete [Lau06].

Le and Tuy in 2007 generalized tree-root results further by giving a linear time recognition of roots of block graphs [LT07].

Extending to the broader generalizing of graph squares, Mukhopadhyay in 1967 provided necessary and sufficient conditions for a connected graph to have a generalized square root [Muk67], whilst Geller in 1970 likewise found necessary and sufficient conditions for a digraph to have a square root [Gel70].

It was Motwani and Sudan in 1994 who established the complexity of the square problem for the generalized undirected case and showed that the problem of determining if a given graph has a square root is a NP-complete problem via a polynomial reduction to 3-CNF Not-All-Equal SAT [MS94].

Their work led to joint work by Farzad, Le, Tuy, Lau and Karimi in 2009 and 2012, in which they established a partial dichotomy theorem for the complexity of determining if a given undirected graph of girth n has a square root [Far+09] which was informally completed under [FK12]. In 2013, Cochefert and others gave an algorithm for constructing a square root with minimal number of edges from undirected graphs with max degree ≤ 6 in polynomial time and an algorithm for constructing maximal edge square roots in exponential time [Coc+13].

In recent years, Duffy provided a full classification for the undirected square of oriented graphs satisfying $\max \text{degree} \leq 3$ and $\text{girth} \geq 4$ [Duf25].

1.3 Definitions

Much of the existing research on graph square roots has centered on characterizing when a graph admits such a root and on the complexity of reconstructing H from G . In contrast, rather than studying the construction problem directly, we analyze the relationship between the adjacency matrix A_H of the root and the adjacency matrix A_G of the square, and show that the problem can be completely categorized as an integer program.

We then examine the relaxed linear program which can be evaluated in polynomial time using interior point method, and integrate existing literature on special root structures as a stepping-stone towards generating a true polynomial time algorithm.

Prior, to this we first standardize the definitions that will be used for the rest of this thesis.

1.3.1 Complexity Theory

Complexity theory studies the classification of computational problems and explores the amount of resources required to solve certain problems. The critical resource that is measured is running time and is standardized across various different problems by using

Definition 1.3.1 (Big- O Notation). *Let $f, g : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$. The **Big- O Notation** $f(n) = O(g(n))$ is written if there exist constants $c > 0$ and $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$*

$$f(n) \leq cg(n).$$

Using this notation, can classify algorithms based on their runtime.

Definition 1.3.2 (Polynomial-Time Algorithm). *An algorithm is said to run in **polynomial time** if there exists a constant $c \geq 0$ such that its running time on inputs of size n is $O(n^c)$.*

Polynomial-time algorithms are important because they are generally regarded as efficient and computationally “simple”. Consequently, the class of problems that admit polynomial-time algorithms forms the basis for the complexity class **P**.

Definition 1.3.3 (Decision Problem). *A **decision problem** is a computational problem whose output is either **Yes** or **No** for every valid input instance.*

Definition 1.3.4 (Class **P**). *The complexity class **P** consists of all decision problems that can be solved by a deterministic algorithm in polynomial time.*

Unfortunately many important decision problems are not known to admit such polynomial-time algorithms and hence need additional classification on their “hardness”. For these problems, despite not have an efficient algorithm for finding a solution, a proposed solution can be verified efficiently.

Definition 1.3.5 (Certificate). *Let L be a decision problem. A **certificate** for an instance x of L is a finite string c that provides evidence that x is a **Yes**-instance of L . A certificate is said to be **polynomially verifiable** if there exists a deterministic algorithm that for given input (x, c) , can determine whether c is a valid certificate for x in time polynomial in $|x|$.*

Definition 1.3.6 (Class **NP**). *The complexity class **NP** consists of all decision problems for which every **Yes**-instance admits a certificate that is polynomially verifiable.*

Since every decision problem in $\mathbf{P}$ can be solved in polynomial time, a solution can also be verified in polynomial time,

$$\mathbf{P} \subseteq \mathbf{NP}.$$

Whether this inclusion is strict remains one of the most important open questions in theoretical computer science - the $\mathbf{P}$ vs $\mathbf{NP}$ problem. As $\mathbf{NP}$ is a large space, the relative difficulty of problems in $\mathbf{NP}$ is compared by using reduction.

Definition 1.3.7 (Polynomial-Time Reduction). *Let A and B be decision problems. A **polynomial-time reduction** from A to B , denoted $A \leq_p B$, is a polynomial-time computable function f such that, for every instance x of A ,*

$$x \in A \iff f(x) \in B.$$

Definition 1.3.8 (Class $\mathbf{NP}$ -Hard). *A decision problem B is **NP-hard** if every problem $A \in \mathbf{NP}$ satisfies $A \leq_p B$. The collection of all **NP-hard** problems is denoted by the class **NP-hard**.*

Definition 1.3.9 (Class $\mathbf{NP}$ -Complete). *A decision problem B is **NP-complete** if $B \in \mathbf{NP}$, and B is **NP-hard**. The collection of all **NP-complete** problems is denoted by the class **NP-complete**.*

1.3.2 Graph Theory

Definition 1.3.10 (Graph).

Definition 1.3.11 (Undirected Graph).

Definition 1.3.12 (Directed Graph).

Definition 1.3.13 (Graph Distance).

Definition 1.3.14 (Graph Root).

Definition 1.3.15 (Graph Square).

Definition 1.3.16 (Adjacency Matrix).

1.3.3 Optimization Problem

Definition 1.3.17 (Integer Program).

Definition 1.3.18 (Linear Program).

Let the *graph root* $H = (V, E_H)$ be a finite, simple, undirected graph, where V is the vertex set with $|V| = n$, and $E_H \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is the edge set. Denote by $d_H(u, v)$ the *graph distance* defined as the number of edges of the shortest path between vertices u and v in H . If no such path exists, $d_H(u, v) = \infty$.

For vertices $u, v \in V$, the *graph distance* $d_H(u, v)$ is the length of a shortest path between u and v in H . If no such path exists, then $d_H(u, v) = \infty$.

The *square* of the graph H , denoted by $H^2 = (V, E_{H^2})$ is the graph on the same vertex set with edge set

$$E_{H^2} = \{uv \mid u \neq v, 1 \leq d_H(u, v) \leq 2\}. \quad (1.1)$$

That is, two distinct vertices are adjacent in H^2 if and only if they are connected by a path of length one or two in H . This is the standard definition of the square of a graph.

Using an alternative construction, let the *graph square* $G = (V, E_G)$ be an undirected graph defined on the same vertex set V . Let $A_H \in \{0, 1\}^{n \times n}$ denote the adjacency matrix of H , where

$$(A_H)_{ij} = \begin{cases} 1, & \text{if } ij \in E_H, \\ 0, & \text{otherwise,} \end{cases}$$

and let $D_H \in \mathbb{R}^{n \times n}$ denote the degree matrix of H , where

$$(D_H)_{ij} = \begin{cases} \deg_H(i), & \text{if } i = j, \\ 0, & \text{otherwise.} \end{cases}$$

Let $\mathbb{1}_{\geq 1} : \mathbb{R}^{n \times n} \rightarrow \{0, 1\}^{n \times n}$ denote the entrywise *indicator function*, defined by

$$\mathbb{1}_{\geq 1}(M)_{ij} = \begin{cases} 1, & \text{if } M_{ij} \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Define the adjacency matrix of G using the adjacency and degree matrices of H

$$A_G = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H). \tag{1.2}$$

Chapter 2

Algorithmic Approaches

In this chapter we explore different methodologies that have been used to show certain properties of the graph root and the graph square. For many restricted cases which are NP, a polynomial time reduction is given and incases which are polynomial, a polynomial time algorithm is given.

We then examine the relaxed linear program which can be evaluated in polynomial time using interior point method, and integrate existing literature on special root structures as a stepping-stone towards generating a true polynomial time algorithm.

2.1 Polynomial Time Algorithm

2.2 NP-Completeness Proofs

2.3 Integer Program Foundation

2.3.1 Background

Lemma 1. *The graph G defined by (1.2) is equivalent to the graph square H^2 defined in (1.1)*

$$ij \in E_{H^2} \iff 1 \leq d_H(i, j) \leq 2. \quad (2.1)$$

Proof. For $i \neq j$, since D_H is diagonal, have $(D_H)_{ij} = 0$ and hence

$$((A_H)^2 - D_H + A_H)_{ij} = (A_H)_{ij}^2 + (A_H)_{ij}.$$

($\Rightarrow$) Suppose $ij \in E_G$ such that $(A_H)_{ij}^2 + (A_H)_{ij} \geq 1$. This implies that at least one of $(A_H)_{ij} = 1$ or $(A_H)_{ij}^2 \geq 1$ must be true. If $(A_H)_{ij} = 1$, then $d_H(i, j) = 1$. If $(A_H)_{ij}^2 \geq 1$, then there exists a vertex k such that $ik, kj \in E_H$, and hence $d_H(i, j) \leq 2$. Therefore $1 \leq d_H(i, j) \leq 2$, and thus $ij \in E_{H^2}$.

($\Leftarrow$) Suppose $ij \in E_{H^2}$ such that $1 \leq d_H(i, j) \leq 2$. D_H is diagonal such that $(D_H)_{ij} = 0$. If $d_H(i, j) = 1$, then $(A_H)_{ij} = 1$. If $d_H(i, j) = 2$, then there exists a vertex k such that $ik, kj \in E_H$, and hence $(A_H)_{ij}^2 \geq 1$. In either case, $(A_H)_{ij}^2 + (A_H)_{ij} \geq 1$, so $(A_G)_{ij} = 1$, and hence $ij \in E_G$.

For $i = j$, since all graphs are simple, have $(E)_{ii} = 0$, $(A)_{ii} = 0$ and hence

$$((A_H)^2 - D_H + A_H)_{ii} = (A_H)_{ii}^2 - (D_H)_{ii}.$$

$(A_H)_{ii}^2 - (D_H)_{ii} = 0$ is always true since the degree matrix of H always has diagonal values $(D_H)_{ii} = \sum_k x_{ik}$ and each edge from vertex i to its neighbour in H will always create a loop

to itself in the square such that $(A_H)_{ii}^2 = \sum_{k=1} x_{ik}x_{ki} = \sum_{k=1} x_{ik}^2$ by the symmetry of adjacency matrix A_H . Since $x_{ik} \in \{0, 1\}$, the square of the boolean is identical to the boolean x_{ik} such that $(A_H)_{ii}^2 = \sum_k x_{ik} = (D_H)_{ii}$ and $(A_G)_{ii} = \mathbb{1}_{\geq 1}((A_H)^2 - D_H + A_H)$ as required.

Combining both gives that $E_G = E_{H^2}$, and so $G = H^2$. □

The formulation of the adjacency matrix of the graph square in Equation (1.2) is very powerful. However, the presence of the indicator function introduces a nonlinearity that prevents the problem from being expressed as a purely quadratic form, thereby limiting the direct applicability of existing results from the quadratic optimization literature.

Lemma 2. *If there are no K_3 triangle subgraphs or any C_4 square subgraphs in the graph root such that H has girth of length 5, then Equation (1.2) can be simplified to become*

$$A_G = ((A_H)^2 - D_H + A_H). \quad (2.2)$$

Proof. Since the graph root has girth of length 5 there are no cycles of length 3 or 4 in H . Thus for every pair of vertices $i \neq j \in V$ there is either no cycle that connects i, j or there is a cycle of at least length 5 between them. In all cases $(D_H)_{ij} = 0$. If there is no cycle, then there is at most one path between i, j

- i) If the path is of length 1, then $(A_H)_{ij} = 1, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0 - 0 + 1 = 1$.
- ii) If the path is of length 2, then $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 1 \implies (A_G)_{ij} = 1 - 0 - 0 = 1$.
- iii) If the path is of length ≥ 3 or a path does not exist, then $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0$.

If there is a cycle of length ≥ 5 between i, j such that there are two paths between i, j

- i) If one path is of length 1, then the other is of length ≥ 4 and hence $(A_H)_{ij} = 1, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0 - 0 + 1 = 1$.
- ii) If one path is of length 2, then the other is of length ≥ 3 and hence $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 1 \implies (A_G)_{ij} = 1 - 0 - 0 = 1$.
- iii) If the paths are both of length ≥ 3 , then $(A_H)_{ij} = 0, (A_H)_{ij}^2 = 0 \implies (A_G)_{ij} = 0$.

If $i = j$, then $(A_H)_{ii} = 0, (D_H)_{ii} = \deg_H(i)$, and $(A_H)_{ii}^2 = \deg_H(i)$. Thus,

$$(A_H)_{ii}^2 - (D_H)_{ii} + (A_H)_{ii} = \deg_H(i) - \deg_H(i) + 0 = 0 = (A_G)_{ii}.$$

□

2.4 Linear Program Relaxation

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